



ΥΠΟΛΟΓΙΣΤΙΚΗ ΝΟΗΜΟΣΥΝΗ

Εργαστηριακές Ασκήσεις

ΓΕΡΑΣΙΜΟΣ ΧΑΝΤΖΗΣ ΑΜ: 3360
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ΝΙΚΟΛΕΤΑ ΤΟΥΡΟΥΝΟΓΛΟΥ ΑΜ:5106

K-Means Αλγόριθμος:

Η άσκηση αφορά την ομαδοποίηση ενός συνόλου δεδομένων (ΣΔΟ) σε M ομάδες χρησιμοποιώντας τον αλγόριθμο K-Means. Θέλουμε να εκτελεστεί ο αλγόριθμος για διάφορες τιμές του M (4, 6, 8, 10, 12), να βρεθεί η βέλτιστη λύση με βάση το σφάλμα ομαδοποίησης, και να αποθηκευτούν τα αποτελέσματα.

Το πρόγραμμα, αρχικά, διαβάζει δεδομένα από το αρχείο Paradeigmata.txt. Κάθε σημείο χι αποτελείται από δύο συντεταγμένες x1 και x2. Τα δεδομένα αποθηκεύονται σε έναν πίνακα αντικειμένων τύπου Point.

```
72     private static void readFile() throws FileNotFoundException {
73         Scanner sc = new Scanner(new File(PARADEIGMATA_FILENAME));
74         for (int i = 0; i < N; i++) {
75             float x1 = sc.nextFloat();
76             float x2 = sc.nextFloat();
77             shmeia[i] = new Point(x1, x2);
78         }
79         sc.close();
80     }
```

Στη συνέχεια, γίνεται αρχικοποίηση κέντρων επιλέγοντας τυχαία ένα από τα σημεία του συνόλου δεδομένων. Αυτή η διαδικασία μας εξασφαλίζει ότι τα αρχικά κέντρα βρίσκονται εντός του πεδίου των δεδομένων.

```
82     private static void randomCenters() {
83         Random rand = new Random(System.currentTimeMillis());
84         for (int i = 0; i < M; i++) {
85             int randomPoint = rand.nextInt(N);
86             centers[i] = new Point(shmeia[randomPoint].x1, shmeia[randomPoint].x2);
87         }
88     }
```

Για κάθε σημείο, υπολογίζεται η Ευκλείδεια απόσταση από κάθε κέντρο. Το σημείο τοποθετείται στην ομάδα του πλησιέστερου κέντρου. Αν υπάρξει αλλαγή ομάδας σε κάποιο σημείο, το πρόγραμμα συνεχίζει την εκτέλεση.

```
90     private static void makeTeams() {
91         for (int i = 0; i < N; i++) {
92             float dx = centers[0].x1 - shmeia[i].x1;
93             float dy = centers[0].x2 - shmeia[i].x2;
94             float minDist = (float) Math.sqrt(dx * dx + dy * dy);
95             int bestTeam = 0;
96
97             for (int j = 1; j < M; j++) {
98                 dx = centers[j].x1 - shmeia[i].x1;
99                 dy = centers[j].x2 - shmeia[i].x2;
100                 float distance = (float) Math.sqrt(dx * dx + dy * dy);
101
102                 if (distance < minDist) {
103                     minDist = distance;
104                     bestTeam = j;
105                 }
106             }
107
108             if (shmeia[i].previousTeam != bestTeam) {
109                 flag = true;
110             }
111             shmeia[i].team = bestTeam;
112             shmeia[i].previousTeam = bestTeam;
113         }
114     }
```

Με βάση τα σημεία που ανήκουν σε κάθε ομάδα, υπολογίζονται οι νέες συντεταγμένες των κέντρων ως ο μέσος όρος των συντεταγμένων των σημείων της ομάδας.

```
118     private static void setNewCenters() {
119         resetCoordinates();
120         sumPointsByTeam();
121         calculateNewCenters();
122     }
123
124     private static void resetCoordinates() {
125         for (int i = 0; i < M; i++) {
126             shmeiaX1[i] = 0.0f;
127             shmeiaX2[i] = 0.0f;
128             teamNumber[i] = 0;
129         }
130     }
131
132     private static void sumPointsByTeam() {
133         for (int i = 0; i < N; i++) {
134             int team = shmeia[i].team;
135             teamNumber[team]++;
136             shmeiaX1[team] += shmeia[i].x1;
137             shmeiaX2[team] += shmeia[i].x2;
138         }
139     }
140
141     private static void calculateNewCenters() {
142         for (int i = 0; i < M; i++) {
143             if (teamNumber[i] != 0) {
144                 centers[i].x1 = shmeiaX1[i] / teamNumber[i];
145                 centers[i].x2 = shmeiaX2[i] / teamNumber[i];
146             }
147         }
148     }
```

Το πρόγραμμα ελέγχει αν τα κέντρα έχουν αλλάξει. Αν δεν υπάρξει αλλαγή, η διαδικασία τερματίζεται.

```
168     private static boolean hasCentersChanged(Point[] oldCenters, Point[] newCenters) {
169         for (int i = 0; i < oldCenters.length; i++) {
170             float dx = oldCenters[i].x1 - newCenters[i].x1;
171             float dy = oldCenters[i].x2 - newCenters[i].x2;
172             float distanceSquared = dx * dx + dy * dy;
173             if (distanceSquared != 0) {
174                 return true;
175             }
176         }
177         return false;
178     }
```

Το σφάλμα υπολογίζεται ως το άθροισμα των Ευκλείδειων αποστάσεων κάθε σημείου από το κέντρο της ομάδας του.

```
180     private static void calculateError() {
181         sfalma = 0.0f;
182         for (int i = 0; i < N; i++) {
183             float dx = centers[shmeia[i].team].x1 - shmeia[i].x1;
184             float dy = centers[shmeia[i].team].x2 - shmeia[i].x2;
185             sfalma += (float) Math.sqrt(dx * dx + dy * dy);
186         }
187     }
```

Για κάθε τιμή του M, εκτελούνται 20 τρεξίματα με διαφορετικά αρχικά κέντρα. Καταγράφεται το σφάλμα για κάθε τρέξιμο, και αποθηκεύεται η λύση με το μικρότερο σφάλμα.

```
43     for (int run = 1; run <= 20; run++) {
44         randomCenters();
45         int times = 0;
46         while (true) {
47             Point[] oldCenters = copyCenters(centers);
48             makeTeams();
49             setNewCenters();
50             times++;
51
52             if (!hasCentersChanged(oldCenters, centers)) {
53                 break;
54             }
55             //System.out.printf("Changed Centers %d times\n", times);
56         }
57         calculateError();
58         float currentError = sfalma / N;
59         System.out.printf("Run #d, Centers changed %d times, Clustering Error = %f\n",
60             run, times, currentError);
61         if (currentError < minError) {
62             minError = currentError;
63             finalCenters = copyCenters(centers);
64             for (int i = 0; i < N; i++) {
65                 finalTeams[i] = shmeia[i].team;
66             }
67         }
68     }
69     System.out.printf("Minimum Error = %f\n", minError);
70     writeCentersToFile(finalCenters);
71 }
```

Τα κέντρα της βέλτιστης λύσης και το ελάχιστο σφάλμα αποθηκεύονται σε αρχείο (Centers.txt).

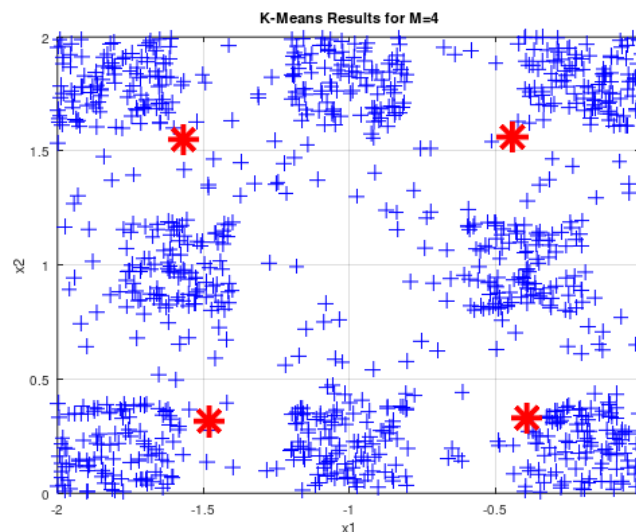
```
160     private static void writeCentersToFile(Point[] centers) throws FileNotFoundException {
161         PrintWriter pw = new PrintWriter(CENTERS_FILENAME);
162         for (int i = 0; i < M; i++) {
163             pw.printf("%f %f\n", centers[i].x1, centers[i].x2);
164         }
165         pw.close();
166     }
```

Αποτελέσματα - Γραφική Απεικόνιση

Για κάθε M, δημιουργήθηκε διάγραμμα που απεικονίζει τα σημεία και τα κέντρα των ομάδων. Τα σημεία του συνόλου δεδομένων απεικονίζονται ως '+' και τα κέντρα των ομάδων απεικονίζονται ως '*'.

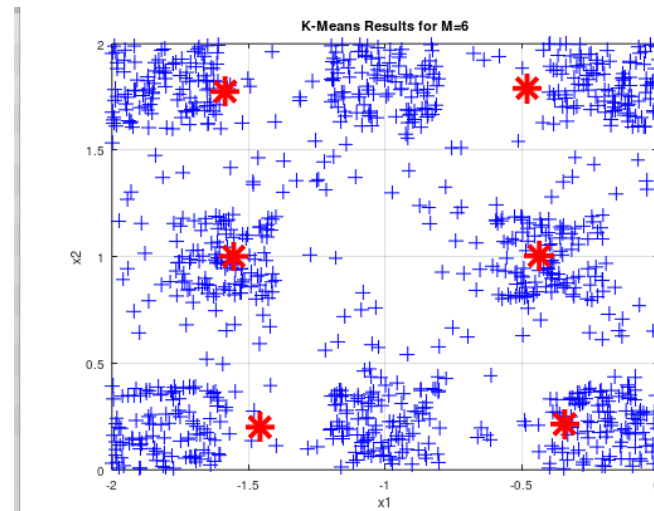
M=4:

```
M = 4
Run #1, Centers changed 5 times, Clustering Error = 0.452264
Run #2, Centers changed 9 times, Clustering Error = 0.412460
Run #3, Centers changed 5 times, Clustering Error = 0.452264
Run #4, Centers changed 9 times, Clustering Error = 0.452264
Run #5, Centers changed 15 times, Clustering Error = 0.418633
Run #6, Centers changed 15 times, Clustering Error = 0.411969
Run #7, Centers changed 23 times, Clustering Error = 0.411969
Run #8, Centers changed 16 times, Clustering Error = 0.418933
Run #9, Centers changed 17 times, Clustering Error = 0.416934
Run #10, Centers changed 11 times, Clustering Error = 0.414732
Run #11, Centers changed 29 times, Clustering Error = 0.411665
Run #12, Centers changed 13 times, Clustering Error = 0.414285
Run #13, Centers changed 10 times, Clustering Error = 0.417015
Run #14, Centers changed 10 times, Clustering Error = 0.451339
Run #15, Centers changed 12 times, Clustering Error = 0.414468
Run #16, Centers changed 14 times, Clustering Error = 0.414285
Run #17, Centers changed 14 times, Clustering Error = 0.414732
Run #18, Centers changed 18 times, Clustering Error = 0.416934
Run #19, Centers changed 15 times, Clustering Error = 0.418933
Run #20, Centers changed 14 times, Clustering Error = 0.418933
Minimum Error = 0.411665
```



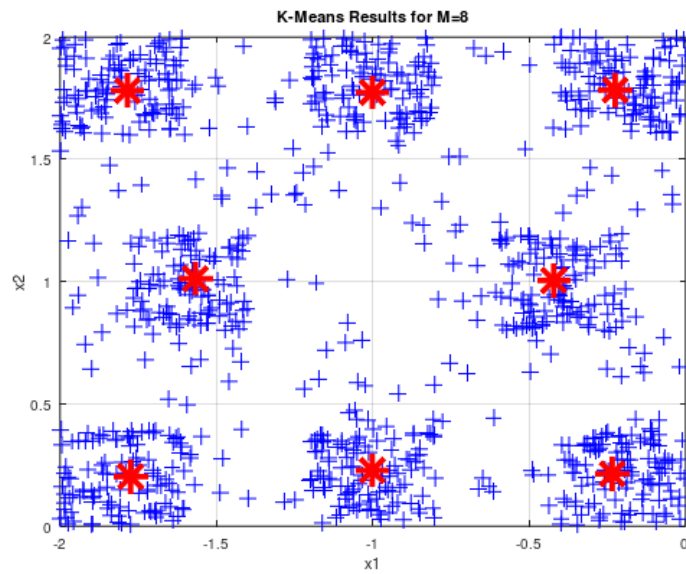
M=6:

```
M = 6
Run #1, Centers changed 16 times, Clustering Error = 0.306148
Run #2, Centers changed 25 times, Clustering Error = 0.299251
Run #3, Centers changed 12 times, Clustering Error = 0.298514
Run #4, Centers changed 21 times, Clustering Error = 0.303493
Run #5, Centers changed 27 times, Clustering Error = 0.303493
Run #6, Centers changed 20 times, Clustering Error = 0.300939
Run #7, Centers changed 12 times, Clustering Error = 0.302921
Run #8, Centers changed 8 times, Clustering Error = 0.298369
Run #9, Centers changed 18 times, Clustering Error = 0.300956
Run #10, Centers changed 22 times, Clustering Error = 0.300956
Run #11, Centers changed 14 times, Clustering Error = 0.300917
Run #12, Centers changed 18 times, Clustering Error = 0.299618
Run #13, Centers changed 14 times, Clustering Error = 0.303581
Run #14, Centers changed 10 times, Clustering Error = 0.298514
Run #15, Centers changed 10 times, Clustering Error = 0.298514
Run #16, Centers changed 14 times, Clustering Error = 0.300939
Run #17, Centers changed 16 times, Clustering Error = 0.303483
Run #18, Centers changed 16 times, Clustering Error = 0.303483
Run #19, Centers changed 11 times, Clustering Error = 0.300180
Run #20, Centers changed 16 times, Clustering Error = 0.298087
Minimum Error = 0.298087
```



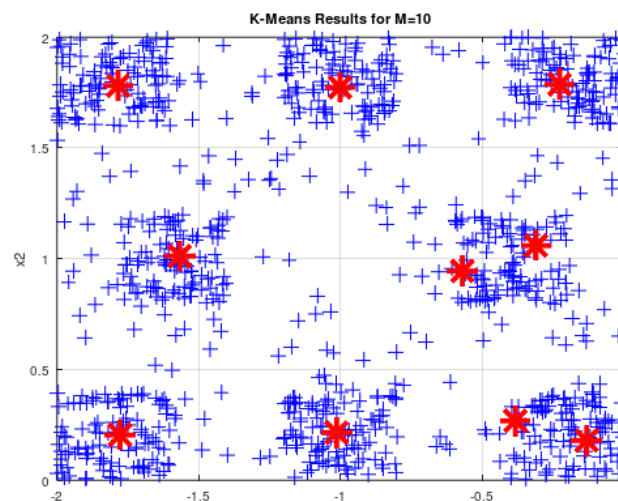
M=8:

```
M = 8
Run #1, Centers changed 14 times, Clustering Error = 0.237775
Run #2, Centers changed 14 times, Clustering Error = 0.235135
Run #3, Centers changed 12 times, Clustering Error = 0.242603
Run #4, Centers changed 5 times, Clustering Error = 0.185285
Run #5, Centers changed 10 times, Clustering Error = 0.239707
Run #6, Centers changed 10 times, Clustering Error = 0.241185
Run #7, Centers changed 30 times, Clustering Error = 0.241113
Run #8, Centers changed 32 times, Clustering Error = 0.239624
Run #9, Centers changed 10 times, Clustering Error = 0.185322
Run #10, Centers changed 13 times, Clustering Error = 0.237564
Run #11, Centers changed 22 times, Clustering Error = 0.242651
Run #12, Centers changed 15 times, Clustering Error = 0.239890
Run #13, Centers changed 11 times, Clustering Error = 0.241111
Run #14, Centers changed 12 times, Clustering Error = 0.185322
Run #15, Centers changed 17 times, Clustering Error = 0.237801
Run #16, Centers changed 20 times, Clustering Error = 0.237469
Run #17, Centers changed 12 times, Clustering Error = 0.236346
Run #18, Centers changed 17 times, Clustering Error = 0.293310
Run #19, Centers changed 16 times, Clustering Error = 0.290989
Run #20, Centers changed 23 times, Clustering Error = 0.237380
Minimum Error = 0.185285
```



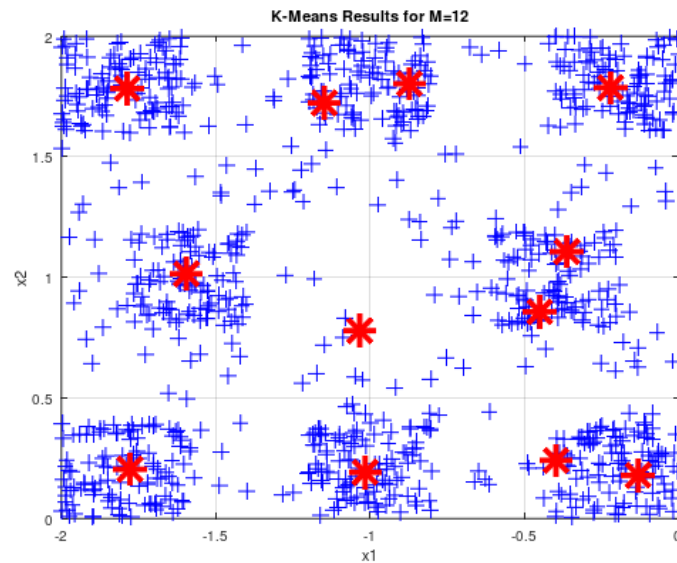
M=10:

```
M = 10
Run #1, Centers changed 14 times, Clustering Error = 0.175122
Run #2, Centers changed 9 times, Clustering Error = 0.176680
Run #3, Centers changed 15 times, Clustering Error = 0.174337
Run #4, Centers changed 12 times, Clustering Error = 0.225500
Run #5, Centers changed 13 times, Clustering Error = 0.177021
Run #6, Centers changed 11 times, Clustering Error = 0.174725
Run #7, Centers changed 16 times, Clustering Error = 0.174075
Run #8, Centers changed 11 times, Clustering Error = 0.174244
Run #9, Centers changed 14 times, Clustering Error = 0.174400
Run #10, Centers changed 8 times, Clustering Error = 0.174404
Run #11, Centers changed 8 times, Clustering Error = 0.174404
Run #12, Centers changed 17 times, Clustering Error = 0.226982
Run #13, Centers changed 7 times, Clustering Error = 0.174226
Run #14, Centers changed 7 times, Clustering Error = 0.174226
Run #15, Centers changed 18 times, Clustering Error = 0.174297
Run #16, Centers changed 8 times, Clustering Error = 0.176207
Run #17, Centers changed 8 times, Clustering Error = 0.176207
Run #18, Centers changed 17 times, Clustering Error = 0.174389
Run #19, Centers changed 10 times, Clustering Error = 0.174860
Run #20, Centers changed 18 times, Clustering Error = 0.174260
Minimum Error = 0.174075
```



M=12:

```
C:\Users\gkara\Desktop\5th year\winter\ypologistiki noimosini>java KMeans.java
M = 12
Run #1, Centers changed 8 times, Clustering Error = 0.171458
Run #2, Centers changed 13 times, Clustering Error = 0.166036
Run #3, Centers changed 25 times, Clustering Error = 0.165637
Run #4, Centers changed 12 times, Clustering Error = 0.163636
Run #5, Centers changed 14 times, Clustering Error = 0.165502
Run #6, Centers changed 18 times, Clustering Error = 0.167913
Run #7, Centers changed 17 times, Clustering Error = 0.164734
Run #8, Centers changed 19 times, Clustering Error = 0.163711
Run #9, Centers changed 13 times, Clustering Error = 0.166481
Run #10, Centers changed 26 times, Clustering Error = 0.163219
Run #11, Centers changed 16 times, Clustering Error = 0.163454
Run #12, Centers changed 36 times, Clustering Error = 0.170002
Run #13, Centers changed 20 times, Clustering Error = 0.166291
Run #14, Centers changed 12 times, Clustering Error = 0.171098
Run #15, Centers changed 15 times, Clustering Error = 0.163997
Run #16, Centers changed 16 times, Clustering Error = 0.166432
Run #17, Centers changed 15 times, Clustering Error = 0.166254
Run #18, Centers changed 14 times, Clustering Error = 0.163293
Run #19, Centers changed 11 times, Clustering Error = 0.166526
Run #20, Centers changed 14 times, Clustering Error = 0.165745
Minimum Error = 0.163219
```



Σφάλμα Ομαδοποίησης

Το σφάλμα μειώνεται με την αύξηση του M , καθώς περισσότερα κέντρα επιτρέπουν καλύτερη προσέγγιση των σημείων. Η σχέση του σφάλματος με τον αριθμό των ομάδων M παρουσιάζεται γραφικά.

